Lie Theory: Chapter 2 Homework

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1. Let I and J be ideals of a Lie algebra L. Prove that $I \cap J$ is a subspace of L.

2. Prove the Isomorphism Theorem (part a).

3. Prove the statements on page 14 (just above example 2.4):

(a) Prove that J (as defined on page 14) is an ideal of L.

(b) Prove that J contains I.

4. Suppose L_1 and L_2 are Lie algebras. Let $L := \{(x_1, x_2) : x_i \in L_i\}$ be the direct sum of their underlying vector spaces.

(a) Show that if we define

$$[(x_1, x_2), (y_1, y_2)] := ([x_1, y_1], [x_2, y_2])$$

then L becomes a Lie algebra, the *direct sum* of L_1 and L_2. We denote this by $L = L_1 \oplus L_2$.

(b) Show that if $L = L_1 \oplus L_2$ then $Z(L) = Z(L_1) \oplus Z(L_2)$ and $L' = L_1' \oplus L_2'$.

5. For each pair of the following Lie algebras over $\mathbb{R}$, decide whether or not they are isomorphic:

(a) the Lie algebra $\mathbb{R}^3_\wedge$ with Lie bracket given by the vector product;

(b) the upper triangular 2×2 matrices over $\mathbb{R}$;

(c) the strictly upper triangular 3×3 matrices over $\mathbb{R}$;

(d) $L = \{x \in \mathfrak{gl}(3, \mathbb{R}) : x^t = -x\}$.